

LoadStar-S –

Modular design: flexible in all applications.



The LoadStar-S central battery system reliably supplies power to safety lighting and emergency lighting (230V AC/216 V DC), and automatically monitors itself as well as each individually connected CG-S light (up to 20 per circuit) via the feed-in line. Thanks to the STAR technology, the circuit type of each connected CG-S light can be programmed as desired within a 50-Hz or 60-Hz supply network using the central battery system controller. This means that the same power circuit is used for mixed operation including maintained lighting, switched maintained lighting and non-maintained lighting – all without an additional data cable! The TFT touch display, available in a 4.3" or 7" design with 512 MB of RAM and 4 GB of flash memory, monitors and controls the central battery system in combination with a separate ACU control module. All features of the connected devices and emergency lighting are monitored automatically and any faults reported.

The integrated search function automatically detects all lights connected to the system and addressed during installation. The VisionGuard monitoring software can be accessed via an Ethernet interface.

Overview of device models

LoadStar-S – central battery system

Overview of new LoadStar-S device models

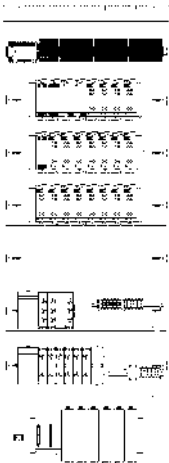


All devices and sub-stations use a modular design. Charging equipment, switching equipment and monitoring equipment form units that operate independently of one another so that the possibility of interactions is excluded. The modular design and the pre-assembled cabinet components guarantee flexible, high-quality implementation. The protection objective of the emergency lighting system is to supply the connected safety lighting system in the event of a general power failure in the primary external power supply. Furthermore, an important function of the emergency lighting system is to ensure the permanent operational capability of all the connected security and LED escape sign luminaires via automatic monitoring.

Depending on the project requirement, the correct device model can be chosen from the LoadStar-S product family.

LoadStar-S 20

for operation with a maximum of 20 SKU.1 CG-S circuit modules with 84 circuit terminals. Up to 6 sub-stations can be supplied with battery power or main power (up to 6 sub-stations 1-phase, up to 2 sub-stations 3-phase).



LoadStar-S 20

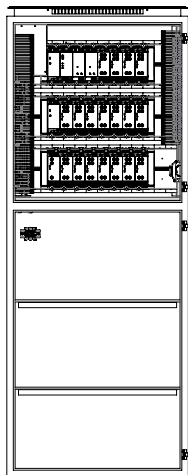
Order details

Model	Included with delivery	Order no.
LoadStar-S 20	Free-standing cabinet with optimized wiring system for central battery system LoadStar-S 20, equipped with battery control module (BCM.1), advanced control unit (ACU) and power supply unit (PSU). For installing a 4.3" or 7" TFT touch display. With extra space for adding a maximum of 84 final circuits, or a maximum of 20 variable circuit boards. Please note! The CM charging modules and the TFT touch display are not part of the cabinet module.	40071362670

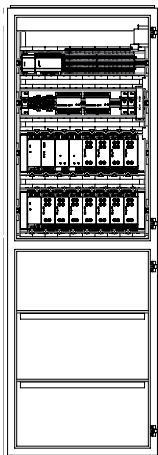
LoadStar-S 20C6, LoadStar-S 12C4

for operation with a maximum of 12 or 20 SKU.1 CG-S circuit modules.

Note: The graphics are not to scale.



LoadStar-S 20C6



LoadStar-S 12C4

Order details

Model	Included with delivery	Order no.
LoadStar-S 20C6	Compact cabinet for central battery system LoadStar-S 20C6, equipped with battery control module (BCM.1), advanced control unit (ACU) and power supply unit (PSU). For installing a 4.3" or 7" TFT touch display. With extra space for adding a maximum of 68 final circuits, or a maximum of 20 variable circuit boards. Please note! The CM charging modules and the TFT touch display are not part of the cabinet module.	40071362671
LoadStar-S12C4	Compact cabinet for central battery system LoadStar-S 12C4, equipped with battery control module (BCM.1), advanced control unit (ACU) and power supply unit (PSU). For installing a 4.3" or 7" TFT touch display. With extra space for adding a maximum of 48 final circuits, or a maximum of 12 variable circuit boards. Please note! The CM charging modules and the TFT touch display are not part of the cabinet module.	40071362672

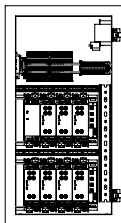
Overview of device models

LoadStar-S – central battery system

LoadStar-S US 7

for operation with a maximum of 7 SKU.1 CG-S circuit modules.
Charging technology for connected battery emergency power supply not included for these sub-stations; battery and mains power supply takes place via the LoadStar-S system.

Note: The graphics are not to scale.



LoadStar-S US 7

Order details

Model	Included with delivery	Order no.
LoadStar-S US 7	Wall cabinet for LoadStar-S/US-S 7 sub-stations, equipped with advanced control unit (ACU) and power supply unit (PSU). For installing a 4.3" or 7" TFT touch display. With extra space for adding a maximum of 28 final circuits, or a maximum of 7 variable circuit boards. Please note! The TFT touch display is not part of the cabinet module	40071362673

Order details

LoadStar-S – central battery system



Order details

Model	Included with delivery	Order no.
LoadStar-S 20 central battery system	LoadStar-S/20 central battery system including TFT touch display, BCM.1 and PSU, 20 free module slots	40071362670
LoadStar-S LoadStar-S 20C6 central battery system	LoadStar-S 20C6 central battery system including TFT touch display, BCM.1 and PSU, 20 free module slots	40071362671
LoadStar-S 12C4 central battery system	LoadStar-S 12C4 central battery system including TFT touch display, BCM.1 and PSU, 12 free module slots	40071362672
LoadStar-S US 7 sub-station	LoadStar-S US 7 model sub-station including TFT touch display, 7 free module slots	40071362673
LoadStar-S SOU2 sub-station	SOU2 model sub-station including 2 x SOU CG-S 2 x 4A	40071362675
LoadStar-S SOU1 sub-station	SOU1 model sub-station including 1 x SOU CG-S 2 x 4 A	40071362674

Technical data

LoadStar-S – central battery system

LoadStar-S free standing cabinet

LoadStar-S compact cabinets

Model	LoadStar-S 20	LoadStar-S 20C6	LoadStar-S12C4	LoadStar-S4C3
Order code	40071362670	40071362671	40071362672	40071362676
Control section: ACU & HMI	1	1	1	1
PSU	1-2	1-2	1	1
BCM.1	1	1	1	1
SKU.1 CG-S circuit module	0-20 ¹	0-20 ¹	0-12	0-5
CM.1 1.7 A charging module	0-1	0-1	0-2	1
CM 3.4 A charging module	0-6	0-2	–	–
Possible number of 3-PM-IO, 3-PM-IO-INV modules for installation in LoadStar-S	0-4	0-1	0-4	0
Possible number of 3-PM modules for installation in LoadStar-S.	0-4	0-1	0-4	0
Possible number of AC transformers for installation in LoadStar-S	1-2	1-2	1	0
Cabinet design, electric:				
Rated voltage	400/230V	400/230V	230V	230V
Rated frequency	50/60 Hz	50/60 Hz	50/60 Hz	50/60 Hz
Cable placement and grounding system in mains mode / battery mode	TN-C-S/IT	TN-C-S/IT	TN-C-S/IT	TN-C-S/IT
Max. ambient temperature ^a	-5°C to +35°C	-5°C to +35°C	-5°C to +35°C	-5°C to +35°C
Protection class	I	I	I	I
Degree of protection	IP21	IP21	IP21	IP21
Max. rated current (mains) [Σ L1, L2, L3] [A]	80	50	25	12
Max. rated output (mains) [KW]	18.4	14.5	5.8	3.5
Max. rated current (battery) [A]	80	50	25	12
Max. rated output (battery) [KW]	17.3	13.6	5.4	2.6
Three-phase split	yes	–	–	–
Connection diameter for mains and battery feed	50 mm ²	35 mm ²	16 mm ²	16 mm ²
Outlet distributor	0 - 6 outlets DC and AC 1-phase, 0-2 out- puts AC 3-phase	2 outlets	1 outlet	–
Terminal capacity	16 mm ²	35 mm ²	16 mm ²	–
Max. connection diameter final circuit	4 mm ²	4 mm ²	4 mm ²	4 mm ²
Max. number of final circuit connections	84	68	48	20
Cabinet design, mechanical:				
Dimensions H x W x D (mm)	2070 x 800 x 405	2070 x 800 x 600	1800 x 600 x 405	1000 x 600 x 300
Material/design	Sheet steel/free-stand- ing cabinet	Compact cabinet	Compact cabinet	Compact cabinet
Hinge position	Right	Right	Right	Right
Outer finish	Powder coating	Powder coating	Powder coating	Powder coating
Color	RAL 7035	RAL 7035	RAL 7035	RAL 7035
TFT touch display	Yes	Yes	Yes	Yes
Lock mechanism	3 mm double ward key	3mm double ward key	3mm double ward key	3mm double ward key
cable inlets on top	Yes	Yes	Yes	Yes
Cable inlets on bottom	Yes	No	No	No
Base (optional)	100/200	100/200	100/200	–
Weight (w/o battery)	approx. 170 kg	approx. 220 kg	approx. 115 kg	approx. 50 kg
Battery capacity, integrated into:				
Battery cabinet (W x H x D: 2050 x 800 x 400 mm)	23.3-245 Ah	–	–	–
Compact cabinet	–	5.5-89.4 Ah	5.5-23.3 Ah	5.5-16 Ah
Battery rack	23.3-245 Ah	–	–	–

^a1 Optimal ambient temperature battery +20 °C

^a2 If more than 13 SKU.1 CG-S 4 x 1.5 A or 26 SKU.1 CG-S 2 x 3 A/1 x 6 A are fitted, a second PSU module is required. When using two or more PSU modules, it is essential to ensure that all PSU modules are operated on the same module rack, directly next to each other.

LoadStar-S US sub-station

Model	LoadStar-S US 7
Order code:	40071362673
Control section: ACU & HMI	1
PSU	1
SKU.1 CG-S circuit module	0-7
Possible number of 3-PM-IO, 3-PM-IO-INV modules for installation in LoadStar-S	0-1
Possible number of 3-PM modules for installation in LoadStar-S.	0-1
Possible number of AC transformers for installation in LoadStar-S	1
Cabinet design, electric:	
Rated voltage	230V
Rated frequency	50/60 Hz
Cable placement and grounding system in mains/battery mode	TN-C-S/IT
Max. ambient temperature range	-5°C to +35°C
Protection class	I
Degree of protection	IP54
Max. rated current (mains) [$\sum$ L1, L2, L3] [A]	25
Max. rated output (mains) [KW]	6.9
Max. rated current (battery) [A]	25
Max. rated output (battery) [KW]	6.5
Three-phase split	No
Connection diameter for mains and battery feed	16 mm ²
Terminal capacity	–
Max. connection diameter of final circuit	4 mm ²
Max. number of final circuit connections	28
Cabinet design, mechanical:	
Dimensions H x W x D (mm)	600 x 400 x 300
Material/design	Sheet steel/wall cabinet
Hinge position	Right
Outer finish	Powder coating
Color	RAL 7035
TFT touch display	Yes
Lock mechanism	3 mm double ward key
Cable inlets on top	Yes
Cable inlets on bottom	No
Base (optional)	–
Weight (w/o battery)	approx. 42 kg